

Mathematical Statistics Assignment 4

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4.1.3

Let the population be given by $\{1, 2, 3\}$. Let $P(x) = \frac{1}{3}$ for $x = 1, 2, 3$. Take samples of size 3 with replacement.

- (a) Calculate μ and σ^2 .
- (b) Obtain the sampling distribution of the sample mean.
- (c) Obtain the mean and variance of the sample mean.

Solution:

(a) Calculate μ and σ^2

The population is $\{1, 2, 3\}$ with $P(x) = \frac{1}{3}$ for $x = 1, 2, 3$.

Step 1: Calculate the population mean (μ) The population mean is given by:

$$\mu = \frac{1}{N} \sum_{i=1}^N x_i$$

where x_i are the population values and $N = 3$. Substituting the values:

$$\mu = \frac{1}{3}(1 + 2 + 3) = \frac{6}{3} = 2$$

Step 2: Calculate the population variance (σ^2) The population variance is:

$$\sigma^2 = \frac{1}{N} \sum_{i=1}^N (x_i - \mu)^2$$

Substitute $\mu = 2$ and $x_i = \{1, 2, 3\}$:

$$\begin{aligned}\sigma^2 &= \frac{1}{3} [(1 - 2)^2 + (2 - 2)^2 + (3 - 2)^2] \\ \sigma^2 &= \frac{1}{3} [1^2 + 0^2 + 1^2] = \frac{1}{3}(1 + 0 + 1) = \frac{2}{3}\end{aligned}$$

Thus,

$$\mu = 2, \quad \sigma^2 = \frac{2}{3}.$$

(b) Sampling distribution of the sample mean

Step 1: Generate all possible samples of size 3 Since sampling is with replacement, the total number of samples is $3^3 = 27$. The samples are:

$$\{(1, 1, 1), (1, 1, 2), (1, 1, 3), (1, 2, 1), \dots, (3, 3, 3)\}.$$

Step 2: Calculate the sample mean for each sample The sample mean for a sample (x_1, x_2, x_3) is:

$$\bar{x} = \frac{x_1 + x_2 + x_3}{3}.$$

For example: - For $(1, 1, 1)$, $\bar{x} = \frac{1+1+1}{3} = 1$, - For $(1, 1, 2)$, $\bar{x} = \frac{1+1+2}{3} = 1.33$,
- For $(1, 1, 3)$, $\bar{x} = \frac{1+1+3}{3} = 1.67$.

Continue similarly for all 27 samples.

Step 3: Tabulate the sampling distribution Count the frequency of each unique sample mean and calculate their probabilities.

Sample	$\bar{X}$	Sample	$\bar{X}$	Sample	$\bar{X}$
(1, 1, 1)	1	(2, 1, 1)	$1\frac{1}{3}$	(3, 1, 1)	$1\frac{1}{2}$
(1, 1, 2)	$1\frac{1}{3}$	(2, 1, 2)	$1\frac{2}{3}$	(3, 1, 2)	2
(1, 1, 3)	$1\frac{2}{3}$	(2, 1, 3)	2	(3, 2, 1)	2
(1, 2, 1)	$1\frac{1}{3}$	(2, 2, 1)	2	(3, 2, 2)	$2\frac{1}{3}$
(1, 2, 2)	2	(2, 2, 2)	2	(3, 2, 3)	$2\frac{2}{3}$
(1, 2, 3)	$2\frac{1}{3}$	(2, 3, 1)	$2\frac{1}{3}$	(3, 3, 1)	$2\frac{1}{3}$
(1, 3, 1)	2	(2, 3, 2)	$2\frac{2}{3}$	(3, 3, 2)	$2\frac{2}{3}$
(1, 3, 3)	$2\frac{2}{3}$	(2, 3, 3)	3	(3, 3, 3)	3

$\bar{X}$	1	$1\frac{1}{3}$	$1\frac{2}{3}$	2	$2\frac{1}{3}$	$2\frac{2}{3}$	3
$p(\bar{X})$	$\frac{1}{27}$	$\frac{1}{9}$	$\frac{2}{9}$	$\frac{7}{27}$	$\frac{2}{9}$	$\frac{1}{9}$	$\frac{1}{27}$

(c) Mean and variance of the sample mean

Step 1: Mean of the sample mean The mean of the sampling distribution of the sample mean is:

$$E(\bar{X}) = \mu = 2.$$

Step 2: Variance of the sample mean The variance of the sample mean is:

$$\text{Var}(\bar{X}) = \frac{\sigma^2}{n}.$$

Substitute $\sigma^2 = \frac{2}{3}$ and $n = 3$:

$$\text{Var}(\bar{X}) = \frac{\frac{2}{3}}{3} = \frac{2}{9}.$$

Thus,

$$E(\bar{X}) = 2, \quad \text{Var}(\bar{X}) = \frac{2}{9}.$$

4.1.9

The distribution of heights of all students in a large university has a normal distribution with a mean of 66 in. and a standard deviation of 2 in. What is the probability that the mean height of 26 randomly selected students from this university will be more than 67 in.?

Solution:

Let X be the height. $X \sim N(\mu = 66, \sigma^2 = 2^2)$, and

$$\frac{\bar{X} - 66}{\frac{2}{\sqrt{26}}} = Z \sim N(0, 1^2).$$

Then,

$$P(\bar{X} > 170) = P\left(Z > 2\sqrt{26}\right) = 0.$$

4.2.5

Let $X_1, X_2, \dots, X_5$ be a random sample from the normal distribution with mean 55 and variance 223. Let

$$Y = \frac{\sum_{i=1}^5 (X_i - 55)^2}{223}$$

and

$$Z = \frac{\sum_{i=1}^5 (X_i - \bar{X})^2}{223}.$$

- (a) Find the distribution of the random variables Y and Z .
- (b) Are Y and Z independent?
- (c) Find
 - (i) $P(0.554 < Y < 0.831)$, and
 - (ii) $P(0.297 < Z < 0.484)$.

Solution:

(a) Distribution of the random variables Y and Z

Distribution of Y : The term $(X_i - 55)^2$ arises because $X_i \sim N(\mu = 55, \sigma^2 = 223)$. The scaled sum of squared deviations, $\sum_{i=1}^5 (X_i - 55)^2$, follows a chi-square distribution with degrees of freedom equal to the sample size ($n = 5$). Thus:

$$Y = \frac{\sum_{i=1}^5 (X_i - 55)^2}{223} \sim \chi_5^2.$$

Distribution of Z : The term $(X_i - \bar{X})^2$ represents deviations from the sample mean $\bar{X}$. The sum of squared deviations, $\sum_{i=1}^5 (X_i - \bar{X})^2$, follows a chi-square distribution with $n - 1$ degrees of freedom (since one parameter, the mean, is estimated). Thus:

$$Z = \frac{\sum_{i=1}^5 (X_i - \bar{X})^2}{223} \sim \chi_4^2.$$

(b) Are Y and Z independent?

Since Y involves deviations from the population mean (55), and Z involves deviations from the sample mean ($\bar{X}$), their sources of variability are distinct. For a random sample from a normal distribution, the total variance can be partitioned into between-sample and within-sample variance. Thus:

Y and Z are independent.

(c) Calculations

(i) $P(0.62 \leq Y \leq 0.76) = 0.0075$

(ii) $P(0.77 \leq Z \leq 0.95) = 0.0251$

4.2.6

Let X and Y be independent chi-square random variables with 14 and 5 degrees of freedom, respectively. Find:

(a) $P(|X - Y| \leq 11.15)$,

(b) $P(|X - Y| \geq 3.8)$.

Solution:

Let X and Y be independent chi-square random variables with 14 and 5 degrees of freedom, respectively. The random variable $D = X - Y$ can be approximated as normally distributed due to the central limit theorem.

(a) Find $P(|X - Y| \leq 11.15)$

Step 1: Mean and variance of D

$$\mu_X = 14, \quad \sigma_X^2 = 2 \times 14 = 28,$$

$$\mu_Y = 5, \quad \sigma_Y^2 = 2 \times 5 = 10.$$

For $D = X - Y$:

$$\mu_D = \mu_X - \mu_Y = 14 - 5 = 9,$$

$$\sigma_D^2 = \sigma_X^2 + \sigma_Y^2 = 28 + 10 = 38, \quad \sigma_D = \sqrt{38} \approx 6.16.$$

Thus, $D \sim N(9, 38)$.

Step 2: Standardize the inequality

We need:

$$P(|X - Y| \leq 11.15) = P(-11.15 \leq D \leq 11.15).$$

Convert to the standard normal distribution (Z):

$$Z = \frac{D - \mu_D}{\sigma_D}.$$

Substitute:

$$P(-11.15 \leq D \leq 11.15) = P\left(\frac{-11.15 - 9}{6.16} \leq Z \leq \frac{11.15 - 9}{6.16}\right).$$

Simplify:

$$P(-11.15 \leq D \leq 11.15) = P(-3.27 \leq Z \leq 0.35).$$

Step 3: Use the standard normal table

From the standard normal table:

$$P(Z \leq -3.27) \approx 0.0005, \quad P(Z \leq 0.35) \approx 0.6368.$$

Thus:

$$P(-3.27 \leq Z \leq 0.35) = P(Z \leq 0.35) - P(Z \leq -3.27),$$

$$P(-3.27 \leq Z \leq 0.35) = 0.6368 - 0.0005 = 0.6363.$$

Final Answer for (a):

$$P(|X - Y| \leq 11.15) \approx 0.6363.$$

(b) Find $P(|X - Y| \geq 3.8)$

Step 1: Rewrite the inequality

$$P(|X - Y| \geq 3.8) = 1 - P(|X - Y| < 3.8).$$

Step 2: Standardize the inequality

$$P(|X - Y| < 3.8) = P(-3.8 \leq D \leq 3.8).$$

Standardize:

$$P(-3.8 \leq D \leq 3.8) = P\left(\frac{-3.8 - 9}{6.16} \leq Z \leq \frac{3.8 - 9}{6.16}\right).$$

Simplify:

$$P(-3.8 \leq D \leq 3.8) = P(-2.07 \leq Z \leq -0.84).$$

Step 3: Use the standard normal table

From the standard normal table:

$$P(Z \leq -2.07) \approx 0.0192, \quad P(Z \leq -0.84) \approx 0.2005.$$

Thus:

$$P(-2.07 \leq Z \leq -0.84) = P(Z \leq -0.84) - P(Z \leq -2.07),$$

$$P(-2.07 \leq Z \leq -0.84) = 0.2005 - 0.0192 = 0.1813.$$

Step 4: Find $P(|X - Y| \geq 3.8)$

$$P(|X - Y| \geq 3.8) = 1 - P(-3.8 \leq D \leq 3.8),$$

$$P(|X - Y| \geq 3.8) = 1 - 0.1813 = 0.8187.$$

Final Answer for (b):

$$P(|X - Y| \geq 3.8) \approx 0.8187.$$

4.2.15

Find the variance of S^2 , assuming the sample $X_1, X_2, \dots, X_n$ is from $N(\mu, \sigma^2)$.

Solution:

If $X_1, X_2, \dots, X_n$ is from $N(\mu, \sigma^2)$, then, by Theorem 4.2.8, $\frac{(n-1)S^2}{\sigma^2}$ is from $\chi^2_{(n-1)}$.

Then, by Exercise 4.2.13:

$$\text{Var} \left[\frac{(n-1)S^2}{\sigma^2} \right] = 2(n-1).$$

Since $\text{Var}(aX) = a^2 \text{Var}(X)$:

$$\text{Var} \left(\frac{(n-1)^2}{\sigma^4} \text{Var}(S^2) \right) = 2(n-1).$$

Simplifying after multiplying by $\frac{\sigma^4}{(n-1)^2}$, we obtain:

$$\text{Var}(S^2) = \frac{2\sigma^4}{n-1}.$$

4.2.23

Let $X_1, X_2, \dots, X_{n_1}$ be a random sample from a normal population with mean μ_1 and variance σ^2 , and $Y_1, Y_2, \dots, Y_{n_2}$ be a random sample from an independent normal population with mean μ_2 and variance σ^2 . Show that

$$T = \frac{(\bar{X} - \bar{Y}) - (\mu_1 - \mu_2)}{\sqrt{\frac{(n_1-1)S_1^2 + (n_2-1)S_2^2}{n_1+n_2-2} \left(\frac{1}{n_1} + \frac{1}{n_2} \right)}} \sim T_{(n_1+n_2-2)}.$$

Solution:

If $X_1, X_2, \dots, X_{n_1}$ is a random sample from a normal population with mean μ_1 and variance σ^2 , and if $Y_1, Y_2, \dots, Y_{n_2}$ is a random sample from an independent normal population with mean μ_2 and variance σ^2 , then:

$$\bar{X} \sim N\left(\mu_1, \frac{\sigma^2}{n_1}\right), \quad \bar{Y} \sim N\left(\mu_2, \frac{\sigma^2}{n_2}\right), \quad \frac{(n_1-1)S_1^2}{\sigma^2} \sim \chi_{(n_1-1)}^2, \quad \frac{(n_2-1)S_2^2}{\sigma^2} \sim \chi_{(n_2-1)}^2.$$

Then:

$$\bar{X} - \bar{Y} \sim N\left(\mu_1 - \mu_2, \frac{\sigma^2}{n_1} + \frac{\sigma^2}{n_2}\right), \quad \frac{(n_1-1)S_1^2}{\sigma^2} + \frac{(n_2-1)S_2^2}{\sigma^2} \sim \chi_{(n_1+n_2-2)}^2.$$

Thus:

$$\frac{\bar{X} - \bar{Y} - (\mu_1 - \mu_2)}{\sqrt{\frac{\sigma^2}{n_1} + \frac{\sigma^2}{n_2}}} \sim N(0, 1),$$

and:

$$\frac{(n_1-1)S_1^2}{\sigma^2} + \frac{(n_2-1)S_2^2}{\sigma^2} \sim \chi_{(n_1+n_2-2)}^2.$$

Then, since the samples are independent, we have by definition that:

$$\frac{\bar{X} - \bar{Y} - (\mu_1 - \mu_2)}{\sqrt{\frac{\sigma^2}{n_1} + \frac{\sigma^2}{n_2}} \cdot \sqrt{\frac{(n_1-1)S_1^2}{\sigma^2} + \frac{(n_2-1)S_2^2}{\sigma^2}} / \sqrt{n_1 + n_2 - 2}} \sim T_{(n_1+n_2-2)}.$$

This simplifies to:

$$\frac{\bar{X} - \bar{Y} - (\mu_1 - \mu_2)}{\sqrt{\frac{(n_1-1)S_1^2 + (n_2-1)S_2^2}{n_1+n_2-2}} \cdot \left(\frac{1}{n_1} + \frac{1}{n_2}\right)} \sim T_{(n_1+n_2-2)}. \quad \text{Q.E.D.}$$

4.2.26

Let the random variables $X_1, X_2, \dots, X_{10}$ be normally distributed with mean 8 and variance 4. Find a number a such that

$$P\left(\sum_{i=1}^{10} \left(\frac{X_i - 8}{2}\right)^2 \leq a\right) = 0.95.$$

Solution:

Let X and Y be independent chi-square random variables with 14 and 5 degrees of freedom, respectively. The random variable $D = X - Y$ can be approximated as normally distributed due to the central limit theorem.

(a) Find $P(|X - Y| \leq 11.15)$

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$$\mu_X = 14, \quad \sigma_X^2 = 2 \times 14 = 28,$$

$$\mu_Y = 5, \quad \sigma_Y^2 = 2 \times 5 = 10.$$

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$$P(-11.15 \leq D \leq 11.15) = P(-3.27 \leq Z \leq 0.35).$$

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Final Answer for (a):

$$P(|X - Y| \leq 11.15) \approx 0.6363.$$

(b) Find $P(|X - Y| \geq 3.8)$

Step 1: Rewrite the inequality

$$P(|X - Y| \geq 3.8) = 1 - P(|X - Y| < 3.8).$$

Step 2: Standardize the inequality

$$P(|X - Y| < 3.8) = P(-3.8 \leq D \leq 3.8).$$

Standardize:

$$P(-3.8 \leq D \leq 3.8) = P\left(\frac{-3.8 - 9}{6.16} \leq Z \leq \frac{3.8 - 9}{6.16}\right).$$

Simplify:

$$P(-3.8 \leq D \leq 3.8) = P(-2.07 \leq Z \leq -0.84).$$

Step 3: Use the standard normal table

From the standard normal table:

$$P(Z \leq -2.07) \approx 0.0192, \quad P(Z \leq -0.84) \approx 0.2005.$$

Thus:

$$P(-2.07 \leq Z \leq -0.84) = P(Z \leq -0.84) - P(Z \leq -2.07),$$

$$P(-2.07 \leq Z \leq -0.84) = 0.2005 - 0.0192 = 0.1813.$$

Step 4: Find $P(|X - Y| \geq 3.8)$

$$P(|X - Y| \geq 3.8) = 1 - P(-3.8 \leq D \leq 3.8),$$

$$P(|X - Y| \geq 3.8) = 1 - 0.1813 = 0.8187.$$

Final Answer for (b):

$$P(|X - Y| \geq 3.8) \approx 0.8187.$$

4.3.1

The lifetime X of a certain electrical fuse has the following pdf:

$$f(x) = \begin{cases} \frac{1}{10}e^{-x/10}, & x > 0, \\ 0, & \text{otherwise.} \end{cases}$$

Suppose two such fuses are in series and operate independently in a system. Find the pdf of the lifetime Y of the system. (The system will work only if both of the fuses operate.)

Solution:

The probability density function (pdf) of X is:

$$f(x) = \frac{1}{10}e^{-x/10}, \quad x > 0.$$

The cumulative distribution function (CDF) of τ_1 is:

$$F_{\tau_1}(t) = \int_0^t \frac{1}{10}e^{-x/10} dx = -e^{-x/10} \Big|_0^t = 1 - e^{-t/10}.$$

Let Y represent the life length of the system. Since the system operates as long as both fuses work,

$$Y = \min(\tau_1, \tau_2),$$

where τ_1 and τ_2 are independent lifetimes of the two fuses. The CDF of Y is:

$$F_Y(y) = 1 - [1 - F_{\tau_1}(y)]^2.$$

The pdf of Y is given by:

$$f_Y(y) = 2f_{\tau_1}(y)[1 - F_{\tau_1}(y)],$$

which simplifies to:

$$f_Y(y) = \begin{cases} \frac{1}{5}e^{-y/5}, & 0 < y < \infty, \\ 0, & \text{otherwise.} \end{cases}$$

4.3.4

Let $X_1, \dots, X_{10}$ be a random sample from $U[0, 1]$. Find the joint density of Y_2 and Y_7 , where $Y_i, i = 1, 2, \dots, 10$ are order statistics of $X_1, \dots, X_{10}$.

Solution:

Step 1: Order Statistics

The order statistics are defined as the sorted values of the sample:

$$Y_1 \leq Y_2 \leq \dots \leq Y_{10},$$

where Y_2 is the second smallest value and Y_7 is the seventh smallest value.

Step 2: Joint Distribution of Order Statistics

The joint probability density function (PDF) of two order statistics Y_2 and Y_7 for a sample of size $n = 10$ is given by:

$$f_{Y_2, Y_7}(y_2, y_7) = \begin{cases} \frac{n!}{(k-1)!(m-k-1)!(n-m)!} y_2^{k-1} (y_7 - y_2)^{m-k-1} (1 - y_7)^{n-m}, & 0 \leq y_2 \leq y_7 \leq 1, \\ 0, & \text{otherwise.} \end{cases}$$

where:

- $k = 2$ (index of Y_2),
- $m = 7$ (index of Y_7),
- $n = 10$ (sample size).

Step 3: Substitution and Simplification

Substitute $n = 10$, $k = 2$, and $m = 7$ into the formula:

$$f_{Y_2, Y_7}(y_2, y_7) = \begin{cases} \frac{10!}{(2-1)!(7-2-1)!(10-7)!} y_2^1 (y_7 - y_2)^4 (1 - y_7)^3, & 0 \leq y_2 \leq y_7 \leq 1, \\ 0, & \text{otherwise.} \end{cases}$$

Simplify the factorial terms:

$$\frac{10!}{1! \cdot 4! \cdot 3!} = \frac{362880}{1 \cdot 24 \cdot 6} = 2520.$$

Thus:

$$f_{Y_2, Y_7}(y_2, y_7) = \begin{cases} 2520 y_2 (y_7 - y_2)^4 (1 - y_7)^3, & 0 \leq y_2 \leq y_7 \leq 1, \\ 0, & \text{otherwise.} \end{cases}$$

Step 4: Final Answer

The joint density of Y_2 and Y_7 is:

$$f_{Y_2, Y_7}(y_2, y_7) = \begin{cases} 2520 y_2 (y_7 - y_2)^4 (1 - y_7)^3, & 0 \leq y_2 \leq y_7 \leq 1, \\ 0, & \text{otherwise.} \end{cases}$$

4.3.9

Let $X_1, \dots, X_n$ be a random sample from a normal population with mean 10 and variance 4. What is the probability that the largest observation is greater than 10?

Solution:

Let $X_1, \dots, X_n$ be a random sample from $N(10, 4)$. The probability that the largest observation is greater than 10 is:

$$P(Y_n > 10) = 1 - P(Y_n \leq 10).$$

The CDF $F_n(y)$ of Y_n is $[F(y)]^n$, where $F(y)$ is the CDF of X evaluated at y .

Thus:

$$P(Y_n \leq y) = F_n(y) = [F(y)]^n = [P(X \leq y)]^n.$$

For the normal distribution:

$$P(X \leq y) = P\left(Z \leq \frac{y - 10}{2}\right).$$

Substituting:

$$P(Y_n \leq y) = \left[P\left(Z \leq \frac{y - 10}{2}\right) \right]^n.$$

Thus:

$$\begin{aligned} P(Y_n > y) &= 1 - P(Y_n \leq y), \\ P(Y_n > y) &= 1 - \left[P\left(Z \leq \frac{y - 10}{2}\right) \right]^n. \end{aligned}$$

For $y = 10$, this simplifies to:

$$P(Y_n > 10) = 1 - [P(Z \leq 0)]^n,$$

$$P(Y_n > 10) = 1 - (0.5)^n.$$

Projects

Project 1:

1. Generating normal random samples (suppose μ , σ^2 fixed) calculate the median, sample mean, and variance with different sample sizes, for example, $n = 100, 500$, or 1000 and compare with the mean and variance of the population. Draw a table or figure to show them, if necessary.

Solution:

Python Code Generating and Analyze Normal Samples

```
1 import numpy as np
2 import pandas as pd
3 import matplotlib.pyplot as plt
4
5 # Parameters of the population
6 mu = 50 # Population mean
7 sigma2 = 25 # Population variance
8 sigma = np.sqrt(sigma2) # Population standard deviation
9
10 # Sample sizes
11 sample_sizes = [100, 500, 1000]
12
13 # To store results
14 results = []
15
16 # Generate samples and calculate statistics
17 for n in sample_sizes:
18     # Generate a normal random sample
19     sample = np.random.normal(mu, sigma, n)
20
21     # Calculate statistics
22     sample_mean = np.mean(sample)
23     sample_median = np.median(sample)
24     sample_variance = np.var(sample, ddof=1) # Sample variance (
25         unbiased)
26
27     # Append results
28     results.append({
29         "Sample Size": n,
30         "Sample Mean": sample_mean,
31         "Sample Median": sample_median,
32         "Sample Variance": sample_variance
33     })
34
35 # Convert results to a DataFrame
36 df_results = pd.DataFrame(results)
```

```

37 # Display the results in a table
38 print("Sample Statistics Comparison")
39 print(df_results)
40
41 # Plot the sample mean and variance compared to the population
42 plt.figure(figsize=(12, 6))
43
44 # Plot sample mean
45 plt.subplot(1, 2, 1)
46 plt.plot(sample_sizes, df_results["Sample Mean"], marker='o', label
47         ="Sample Mean")
48 plt.axhline(y=mu, color='r', linestyle='--', label="Population Mean
49             ")
50 plt.xlabel("Sample Size")
51 plt.ylabel("Mean")
52 plt.title("Sample Mean vs Population Mean")
53 plt.legend()
54
55 # Plot sample variance
56 plt.subplot(1, 2, 2)
57 plt.plot(sample_sizes, df_results["Sample Variance"], marker='o',
58         label="Sample Variance")
59 plt.axhline(y=sigma2, color='r', linestyle='--', label="Population
60             Variance")
61 plt.xlabel("Sample Size")
62 plt.ylabel("Variance")
63 plt.title("Sample Variance vs Population Variance")
64 plt.legend()
65
66 plt.tight_layout()
67 plt.show()

```

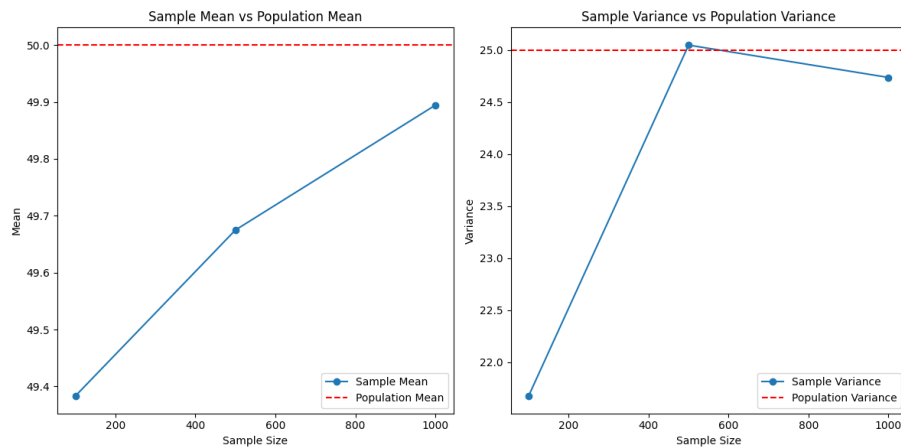


Figure 1: Project 1

```
"D:\Software Application\Python3125\python.exe" "D:\Masters NPU Semester\3rd Semester\Statistics\Homework\Submit\Homework 4\code\Project_1.py"
```

Sample Statistics Comparison				
	Sample Size	Sample Mean	Sample Median	Sample Variance
0	100	49.383424	49.402693	21.674567
1	500	49.674963	49.411294	25.046432
2	1000	49.894409	49.986557	24.735264

Figure 2: Project 1

Project 2:

1. Drawing 10,000 samples, each sample of size 15 (suppose μ, σ^2 fixed)
 Compute the mean of each sample and draw a histogram for the means.
 This will be an approximate sampling distribution of $\bar{X}$ for a fixed sample
 of size 15, and compare with the pdf figure of $\bar{X} \sim N(\mu, \sigma^2/15)$.

Solution:

Python Code for Sampling Distribution of the Sample Mean

```

1 import numpy as np
2 import matplotlib.pyplot as plt
3 from scipy.stats import norm
4
5 # Parameters of the population
6 mu = 50 # Population mean
7 sigma2 = 25 # Population variance
8 sigma = np.sqrt(sigma2) # Population standard deviation
9 sample_size = 15 # Sample size
10 num_samples = 10000 # Number of samples
11
12 # Generate 10,000 samples, each of size 15
13 sample_means = []
14 for _ in range(num_samples):
15     sample = np.random.normal(mu, sigma, sample_size)
16     sample_means.append(np.mean(sample))
17
18 # Convert sample means to a NumPy array
19 sample_means = np.array(sample_means)
20
21 # Compute the parameters of the sampling distribution of the sample
    mean
22 sampling_sigma = sigma / np.sqrt(sample_size) # Standard deviation
    of the sampling distribution
23
24 # Plot histogram of sample means
25 plt.figure(figsize=(10, 6))
26 plt.hist(sample_means, bins=30, density=True, alpha=0.6, color='
    blue', label='Histogram of Sample Means')
27
28 # Plot theoretical normal distribution

```



```

29 x = np.linspace(mu - 4 * sampling_sigma, mu + 4 * sampling_sigma,
30               1000)
31 theoretical_pdf = norm.pdf(x, loc=mu, scale=sampling_sigma)
32 plt.plot(x, theoretical_pdf, 'r-', label='Theoretical PDF ( $N(\mu,$ 
33        $\sigma^2/15)$ '))
34 # Add labels and legend
35 plt.title('Sampling Distribution of the Sample Mean (n=15)')
36 plt.xlabel('Sample Mean')
37 plt.ylabel('Density')
38 plt.legend()
39 plt.grid()
40 plt.show()

```

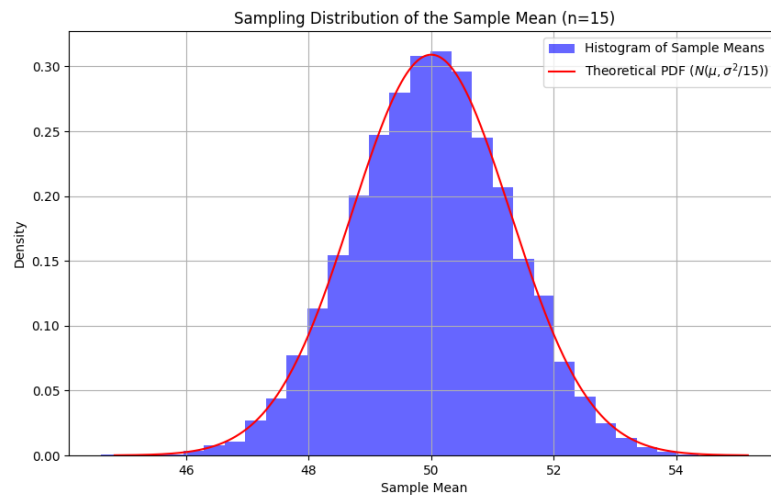


Figure 3: Project 2